LAB WORKSHEET

IDS/IPS Practical Lab

Configure a preconfigured IDS/IPS sensor (Snort 2.9): detect attacks as an IDS, then block them inline as an IPS.

Key files

- `/etc/snort/snort.conf` — set `ipvar HOME_NET 192.168.1.0/24`
- `/etc/snort/rules/local.rules` — your rules (sid 1000001 and up)
- `/var/log/snort/alert`

Commands

```
ip a
sudo nano /etc/snort/snort.conf            (^O save, ^X exit)
sudo snort -T -c /etc/snort/snort.conf     (test the config)
sudo snort -A console -q -c /etc/snort/snort.conf -i eth0                        (IDS)
sudo snort -Q --daq afpacket -i eth0:eth1 -A console -q -c /etc/snort/snort.conf (IPS)
```

Example rules

```
alert icmp $EXTERNAL_NET any -> $HOME_NET any (msg:"ICMP Ping detected"; itype:8; sid:1000001;
rev:1;)
alert tcp $EXTERNAL_NET any -> $HOME_NET 22 (msg:"SSH brute force"; flags:S;
detection_filter:track by_src, count 5, seconds 60; sid:1000002; rev:1;)
drop tcp $EXTERNAL_NET any -> $HOME_NET 80 (msg:"SQL injection"; content:"' OR '1'='1"; nocase;
http_uri; sid:1000003; rev:1;)
```

Guided steps

Check interfaces → fix HOME_NET → test → ICMP rule → run as IDS → ping sweep → SSH brute-force rule → SQLi rule → switch to IPS → drop + run inline → confirm the block and no false positive.